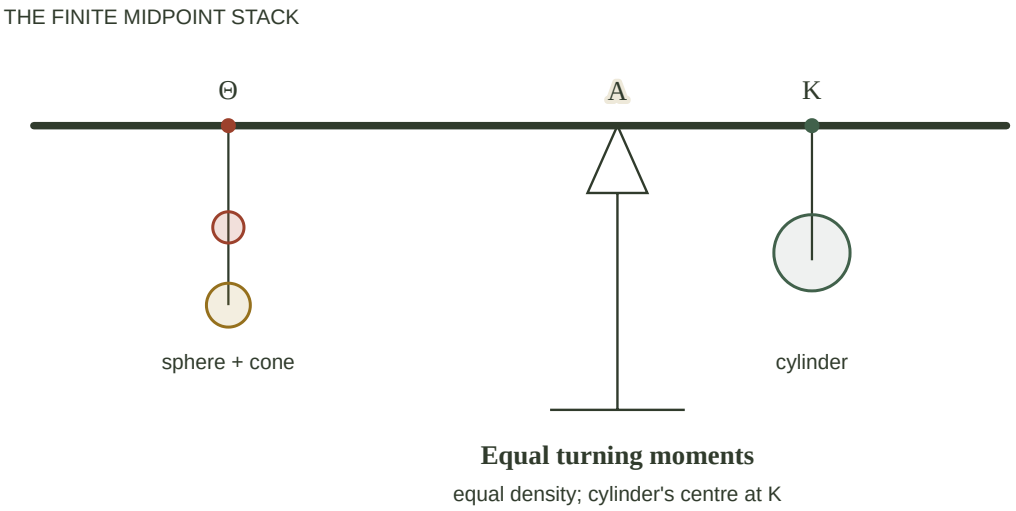
A sphere on the balance.

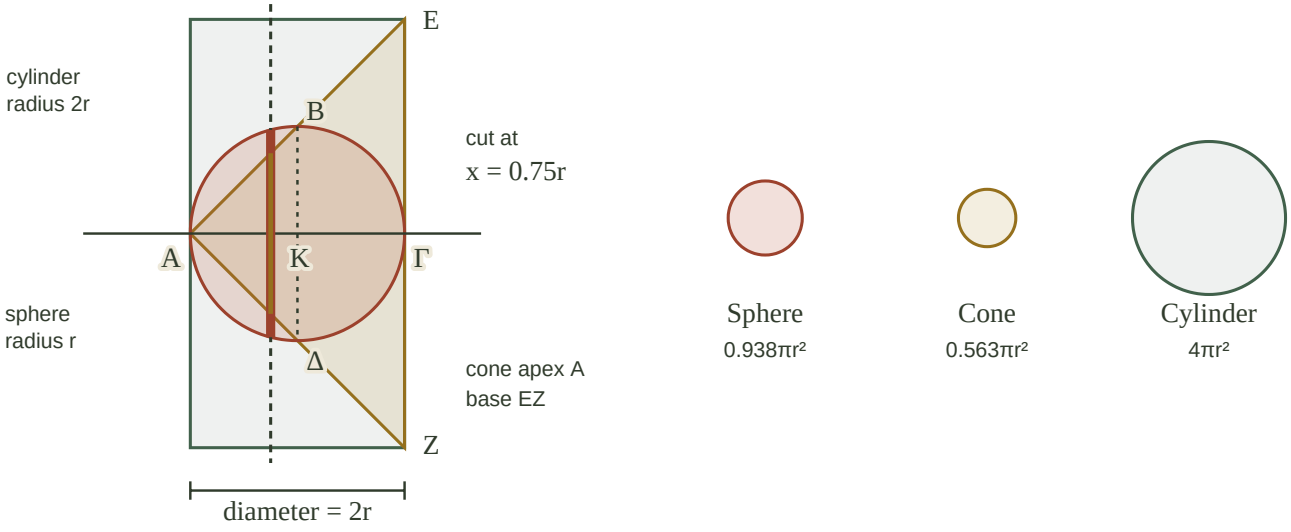
Your retained experiment and all ten selected Greek readings. The Latin renderings are new study aids composed for this workshop, not ancient witnesses.

Cut x = 3/4r. Left arm h = 2r. View: stack. Finite bands: 8. Selected reading: 6.

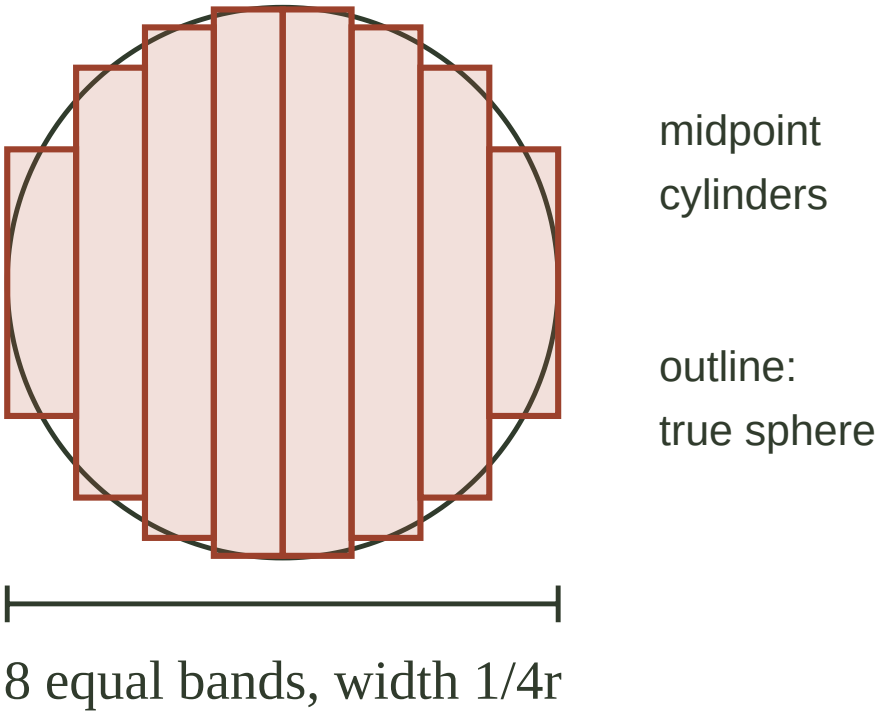
Return online: this experiment and reading. External links need a connection; all illustrations and readings below are embedded.



Left moment: $8\pi r^4$. Right moment: $8\pi r^4$. Deflection indicates sign only.



Finite stacks can conceal opposite errors.



Sphere midpoint estimate: $43/32\pi r^3$; error $+1/96\pi r^3$. Cone estimate: $85/32\pi r^3$; error $-1/96\pi r^3$. Combined estimate: $4\pi r^3$ exactly. Modern enclosing-cylinder bounds: $17/16 \le S/\pi r^3 \le 25/16$.

These bands are a modern approximation. A finite collection of balanced sections does not establish an exact solid volume. Archimedes explicitly distinguishes mechanical discovery from geometrical demonstration.

Keep the cylinders distinct.

Solid	Radius	Height	Volume / πr^3
Auxiliary cylinder C	2r	2r	8
Auxiliary cone K	2r	2r	8/3
Sphere	r	2r	4/3
Enclosing cylinder	r	2r	2
Small cone	r	r	1/3

Balance: S + K = C/2. The cone theorem supplies K = C/3; subtracting leaves S = C/6. This assumes uniform density and a weightless beam. The cone-volume theorem is an input. The source's large auxiliary cylinder has four times the volume of the enclosing cylinder.

Greek display: red angle brackets mark supplied letters, dotted underline marks uncertain readings, italic letters expand abbreviations. Accents, spacing and punctuation are editorial. The early 'twice' is retained alongside the later 'four times'; no new decipherment is claimed.

1 / Arch17r / zArch17r_71v1_19 to zArch17r_71v1_22

A discovery is not yet a proof

Τοῦτο δὴ δι*ὰ* μὲν τῶν νῦν εἰρημέν*ων* οὐκ ἀποδέδεικται, ἔμφασιν δέ τινα πεποίηκε τὸ συμπέρασμα ἀληθὲς εἶναι·

This has not been proved by what has now been said, but it has given some indication that the conclusion is true.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

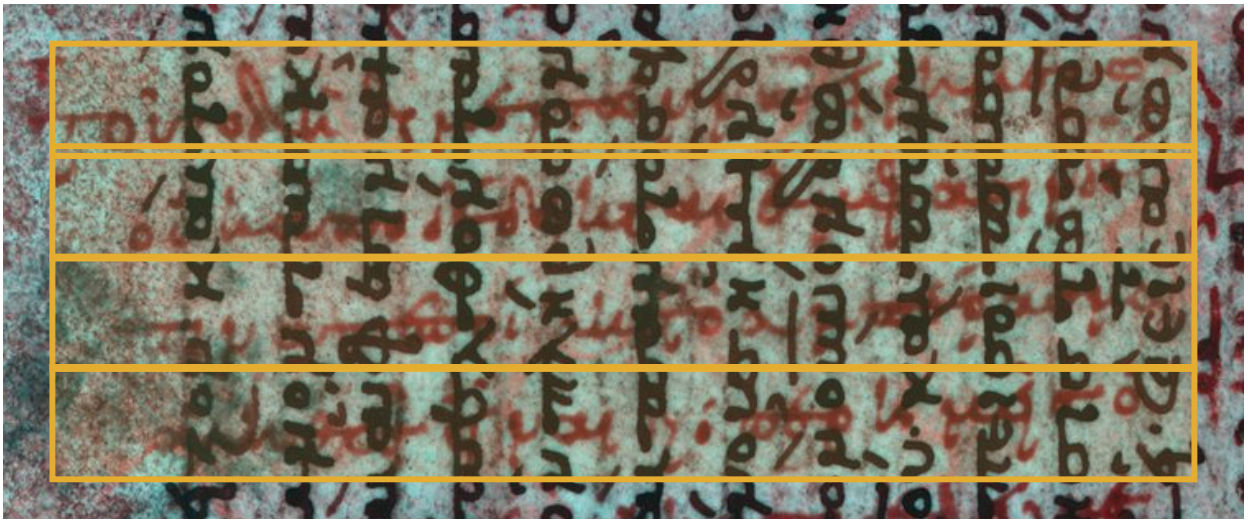
Ex iis quae nunc dicta sunt hoc nondum demonstratum est; indicium tamen quoddam datum est conclusionem veram esse.

οὐκ ἀποδέδεικται

has not been demonstrated · nondum demonstratum est

ἀποδέδεικται is perfect passive: a demonstration has not been established. The negative οὐκ denies the completed proof, not the usefulness of the discovery.

This passage closes the parabola argument before the sphere investigation. In the digital transcription it follows the milestone numbered 2; those divisions follow an edition and are not a new statement that the sphere has already been proved. The letter's introduction also distinguishes mechanical investigation from geometrical demonstration.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

The number that disagrees

ὅτι δὲ πᾶσα σφαῖρα διπλασία ἐστὶν τοῦ κώνου τοῦ βάσιν μὲν ἔχοντος ἴσην τῶι μεγίστωι κύκλωι τῶν ἐν τῆι σφαίραι, ὕψος δὲ ἴσον τῆι ἐκ τ*οῦ* κέντρου τῆς σφαίρας,

That every sphere is twice the cone whose base equals a greatest circle in the sphere and whose height equals the sphere's radius, …

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

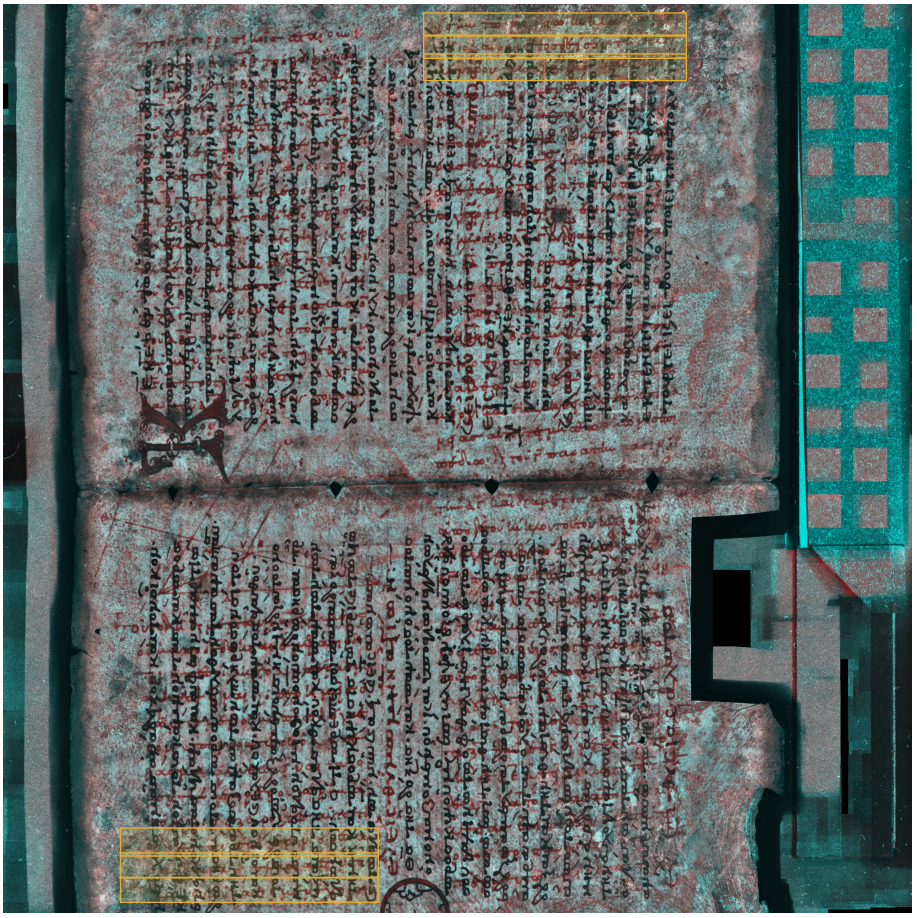
Omnem sphaeram duplam esse coni cuius basis maximo circulo in sphaera aequalis est, altitudo autem radio sphaerae aequalis, …

διπλασία

twice as great · dupla

διπλασία agrees with σφαῖρα, a feminine noun. The cone is expressed in the genitive, τοῦ κώνου: twice the cone. ἡ ἐκ τοῦ κέντρου, literally 'the [line] from the centre', means radius.

The released Netz–Wilson transcription reads διπλασία, 'twice', here; the project's reconstruction of Heiberg's manuscript reading does too. Later in the same argument the transcription reads τετραπλασίων, 'four times'. The workshop keeps the former word rather than silently correcting it. Its geometry and the later conclusion give four times the small cone. This comparison concerns the released readings; it is not a fresh decipherment or a claim about which hand introduced the discrepancy.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

3 / Arch17r / zArch17r_71v2_25 to zArch17r_71v2_28

Imagine the balance

καὶ ἐκβεβλήσθω ἡ ΓΑ, καὶ κείσθω αὐτῇ ἴση ἡ ΑΘ, *καὶ* νοείσθω ὁ ζυγὸς ὁ ΓΘ, μέσον δὲ αὐτοῦ τὸ Α,

Let ΓΑ be extended, and let ΑΘ be laid off equal to it; imagine ΓΘ as a balance, with A at its midpoint, …

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

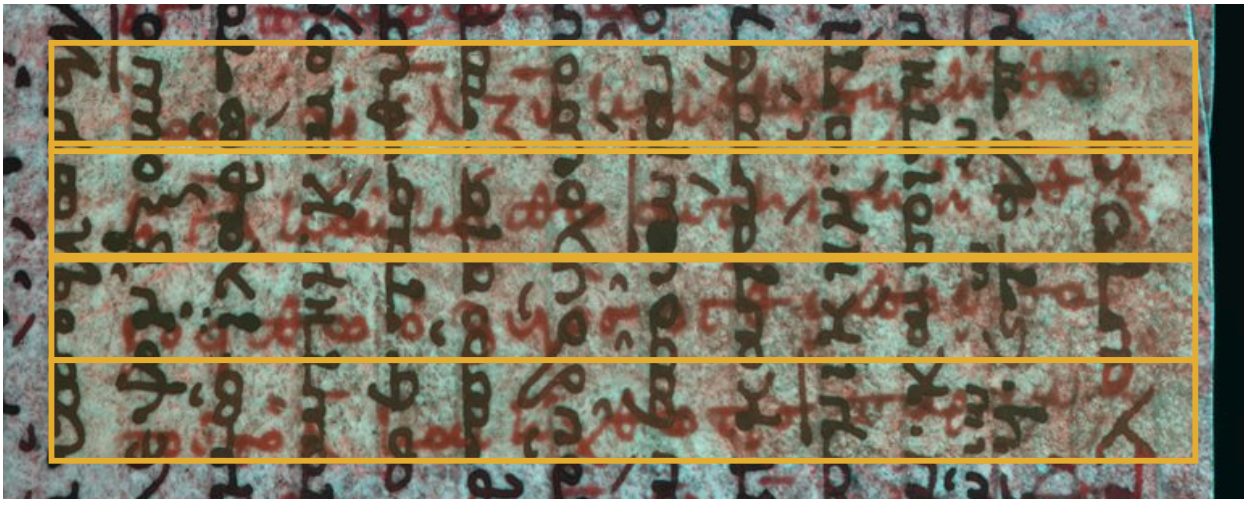
Producatur ΓΑ, et ponatur ΑΘ ipsi aequalis; intellegatur ΓΘ esse libra, cuius medium sit Α, …

νοείσθω ὁ ζυγός

let the balance be conceived · intellegatur libra

νοείσθω is a third-person imperative: 'let it be imagined'. ζυγός here is a balance beam; the project's name, Zygon, recalls the related Greek family of words for a yoke or balance.

ΓΑ is the sphere's diameter, so the fixed left arm ΑΘ is 2r. The fulcrum is at the end A of the sphere, not its centre K. Our small Greek labels follow this transcription: Θ is theta, not the Latin H that appears in some English redrawings.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

4 / Arch17r / zArch17r_71v2_33 to zArch17r_71v2_35

Raise a cutting plane

καὶ ἀπὸ τῆς ΜΝ εὐθείας ἐπίπεδον ἀνεστάτω ὀρθὸν πρὸς τὴν ΑΓ·

And from the straight line MN let a plane be raised perpendicular to AΓ.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

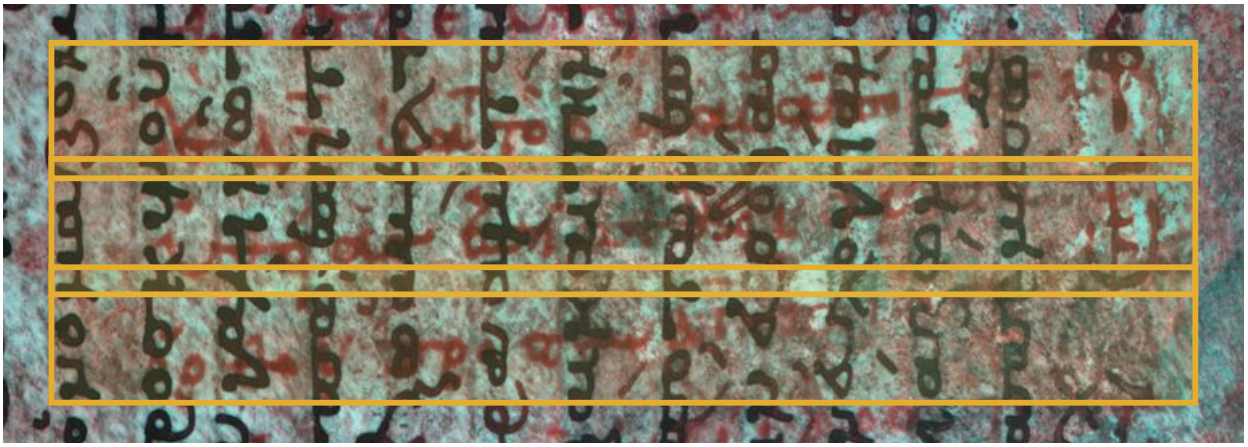
Ex recta MN planum erigatur ad AΓ perpendiculare.

ἐπίπεδον … ὀρθόν

a perpendicular plane · planum perpendiculare

ἀνεστάτω is another construction imperative, 'let it be raised'. ὀρθόν agrees with the neuter ἐπίπεδον. Here it describes the plane's relation to the axis, not the shape of the circle.

The same plane cuts all three solids. The cylinder's circular section is constant, the cone's radius grows with x, and the sphere's radius first grows and then shrinks. Circular areas, not their diameters, become the comparable loads. The source uses geometrical ratios; x, r, π and our decimal controls are modern notation.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

5 / Arch17v / zArch17v_71r1_23 to zArch17v_71r1_26

The inverse ratio

⟨ὡς⟩ ἄρα ἡ ΘΑ *πρὸς* ΑΣ, οὕτως ὁ κύκλος ὁ ἐν τῶι κυλίνδρωι *πρὸς τοὺς* κύκλους τόν τε ἐν τῆι σφαίραι καὶ τὸν ἐν τῶι κώνωι.

Thus, as ΘΑ is to ΑΣ, so is the circle in the cylinder to the circles in the sphere and in the cone.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

Ut igitur ΘΑ ad ΑΣ, ita circulus in cylindro ad circulos in sphaera et in cono.

ὡς … οὕτως

as … so … · ut … ita …

This paired construction states a proportion. πρὸς means 'to' in a ratio. The plural circles on the right are taken together: one from the sphere and one from the cone.

The cylinder section is larger, so it needs the shorter arm. Its area divided by the combined sphere-and-cone area is 2r/x. Inverting that relation gives the lever condition: the pair's area times 2r equals the cylinder's area times x. At x = 0 both moments vanish; there is no meaningful area ratio to divide there.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

6 / Arch17v / zArch17v_71r1_30 to zArch17v_71r1_34

Move the two circles

μετενεχθεῖσιν καὶ τεθεῖσιν οὕτως ἐπὶ τὸ Θ, ὥστε ἑκατέρου αὐτῶν κέντρον εἶναι τοῦ βάρους τὸ Θ, ἰσορροπήσουσι κατὰ τὸ Α σημεῖον.

When they have been transferred and placed at Θ in such a way that Θ is the centre of gravity of each, they will balance about the point A.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

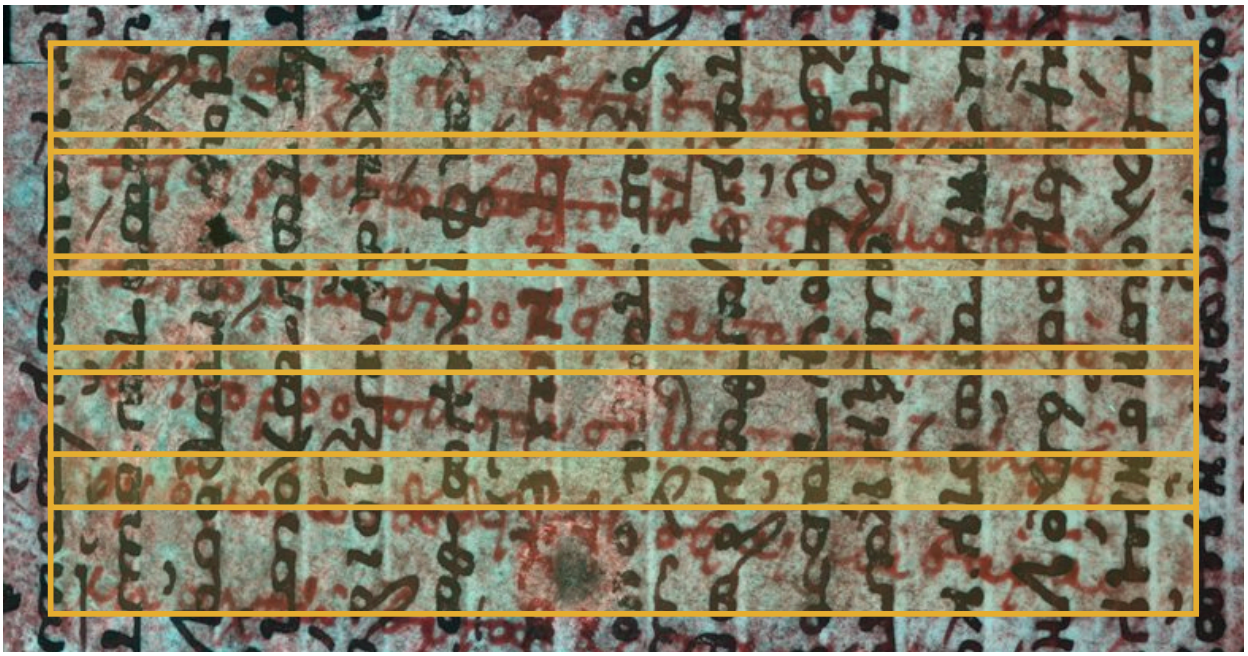
Translati et ad Θ positi, ita ut Θ centrum gravitatis utriusque sit, circa punctum A aequiponderabunt.

μετενεχθεῖσιν

having been transferred · translatis

The passive participle describes the moved circles. ἑκατέρου means 'of each of the two'; both centres are put at Θ. ἰσορροπήσουσι is future, 'they will balance'. The excerpt begins inside a longer sentence.

The cylinder's section stays at its original position Σ. Both other sections are transferred to the same fixed point Θ. Placing them on two neighbouring horizontal hooks would change their moments; the diagram therefore hangs their centres on the same vertical line. Its drawn tilt shows direction only, not a calculated physical angle.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

From circles to solids

συμπληρωθέντος οὖν τοῦ κυλίνδρου ὑπὸ τῶν ληφθέντων κύκλων καὶ τῆς σφαίρας καὶ τοῦ κώνου ἰσορροπήσει ὁ κύλινδρος περὶ τὸ Α σημεῖον αὐτοῦ μένων συναμφοτέροις τῆι τε σφαίραι *καὶ* τῶι κώνωι μετενεχθεῖσι καὶ τεθεῖσι ἐπὶ τοῦ ζυγοῦ *κατὰ* τὸ Θ, ὥστε ἑκατέρου αὐτῶν κέν⟨τ⟩ρ*ον εἶναι* τοῦ βάρους τὸ Θ.

When the cylinder, sphere and cone have been filled by the circles taken, the cylinder, staying where it is, will balance about A with the sphere and cone together, transferred and placed on the balance at Θ so that Θ is the centre of gravity of each.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

Cylindro, sphaera et cono circulis sumptis completis, cylindrus suo loco manens circa A aequiponderabit sphaerae et cono simul ad Θ translatis et in libra positis, ita ut Θ centrum gravitatis utriusque sit.

αὐτοῦ μένων

remaining where it is · suo loco manens

μένων, 'remaining', contrasts with μετενεχθεῖσι, 'transferred'. The opening συμπληρωθέντος is a passive participle: 'when it has been filled'. κέντρον τοῦ βάρους is literally the centre of weight.

This is the crucial mechanical passage from sections to whole solids. Our finite bands have actual thickness and use midpoint radii; they are a modern illustration, not literal zero-thickness circles and not Archimedes' geometrical proof. Their separate sphere and cone estimates retain visible, opposite errors even while their total moments cancel exactly.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

8 / Arch17v / zArch17v_71r2_26 to zArch17v_71r2_30

Subtract the cone

διπλασία δὲ ἡ ΘΑ τῆι ΑΚ· διπλάσιον ἄρα καὶ ὁ κύλινδρος συναμφοτέρου τῆς τε σφαίρας καὶ τοῦ κώνου. αὐτοῦ τε τοῦ κώνου τριπλασίων ἐστί·

ΘA is twice AK; therefore the cylinder is twice the sphere and cone together. It is also three times the cone itself.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

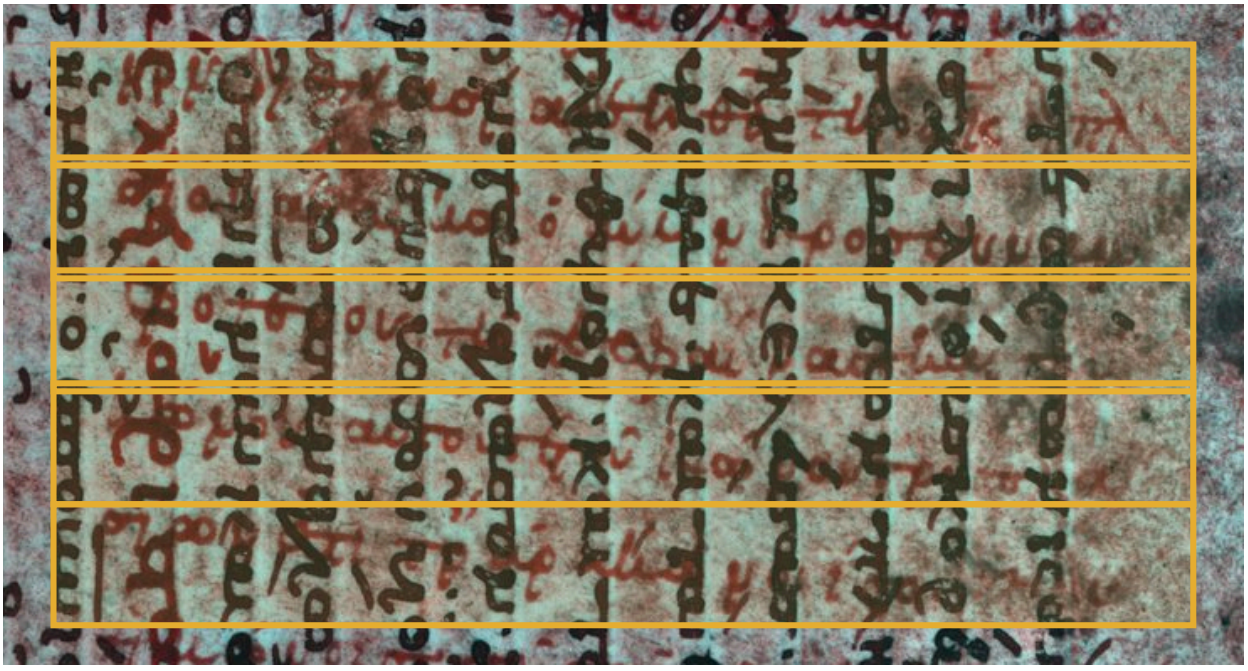
ΘA dupla est AK; cylindrus igitur duplus est sphaerae et coni simul. Idem cylindrus ipsius coni triplus est.

συναμφοτέρου

of both together · utriusque simul

συναμφοτέρου keeps the two magnitudes together. διπλάσιον and τριπλασίων give different comparisons: twice the pair, three times the cone. The ratios are about volumes at the common density.

For the large auxiliary cylinder C, the balance gives S + K = C/2. The cone-volume theorem supplies K = C/3. Subtracting leaves S = C/6. The cone theorem is an input, not something newly established by moving this slider. The original cylinder's centre of gravity is K, one radius from A.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

9 / Arch18r / zArch18r_65r1_05 to zArch18r_65r1_10

Four times the small cone

τετραπλασίων ἄρα ἐστὶν ἡ σφαῖρα, ἧς μέγιστος κύκλος ὁ ΑΒΓΔ, τοῦ κώνου, οὗ κορυφὴ μέν ἐστι τὸ Α σημεῖον, βάσις δὲ ὁ περὶ διάμετρον τὴν ΒΔ κύκλος ὀρθὸς ὢν πρὸς τὸν ΑΓ.

The sphere whose greatest circle is ΑΒΓΔ is therefore four times the cone whose vertex is A and whose base is the circle on diameter ΒΔ, perpendicular to ΑΓ.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

Sphaera igitur cuius maximus circulus est ΑΒΓΔ quadrupla est coni cuius vertex est A, basis autem circulus diametro ΒΔ descriptus, ad ΑΓ perpendicularis.

τετραπλασίων

four times as great · quadrupla

τετρα- supplies 'four'. The Greek comparative form here accompanies ἡ σφαῖρα. The later τετραπλασία in the next selection makes the same fourfold comparison.

This is the small cone of radius r and height r, not the auxiliary cone of radius 2r and height 2r. Doubling every length multiplies volume by eight. The auxiliary cone is twice the sphere; the small cone is one quarter of it. Compare the earlier διπλασία without overwriting either reading.



Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

10 / Arch18r / zArch18r_72v1_29 to zArch18r_72v1_33

The cylinder that encloses the sphere

⟨ἐδ⟩είχθη δὲ τοῦ αὐτοῦ κώνου τετραπλασία οὖσα ἡ σφαῖρα, ἧς μέγιστος μέν ἐστιν ὁ κύκλος ὁ ΑΒΓΔ· ἡμιόλιος ἄρα ὁ κύλινδρος τῆς σφαίρας· ὅπερ ἔδει δειχθῆναι.

The sphere whose greatest circle is ΑΒΓΔ has been shown to be four times that same cone. The cylinder is therefore one and a half times the sphere, which was to be shown.

New Latin study rendering (2026); not an ancient witness or Heiberg translation.

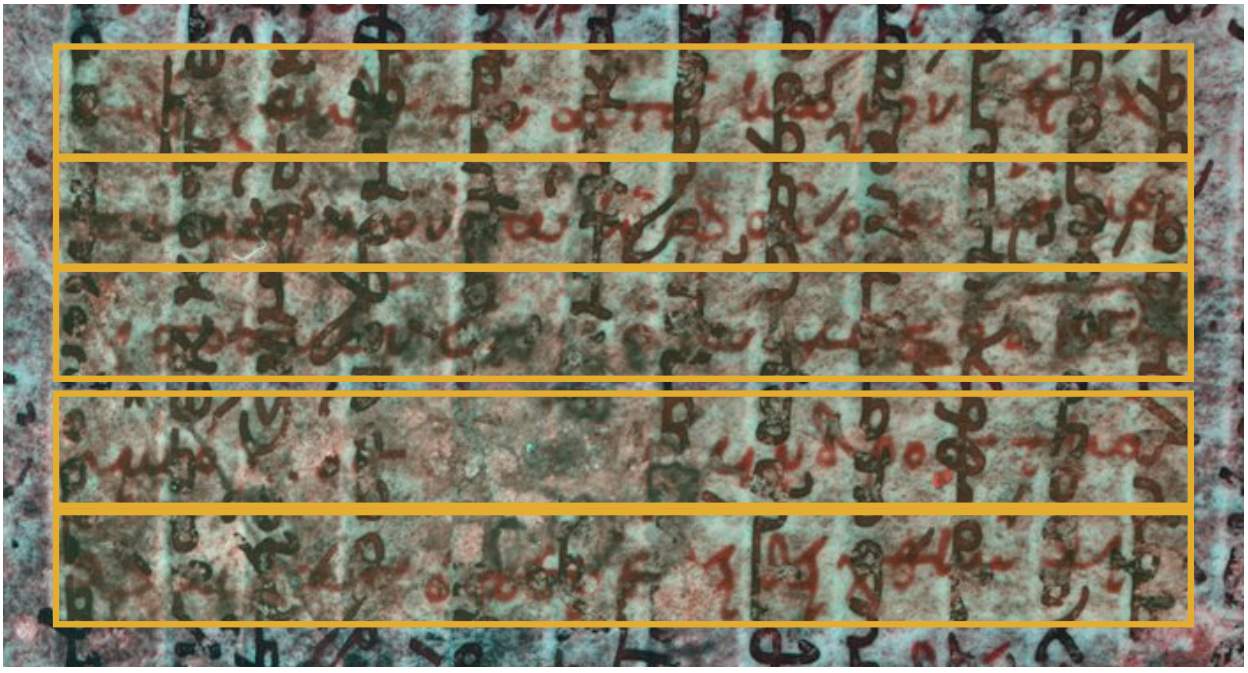
Sphaera cuius maximus circulus est ΑΒΓΔ quadrupla eiusdem coni esse ostensa est; cylindrus igitur sesquialter est sphaerae, quod erat demonstrandum.

ἡμιόλιος

one and a half times · sesquialter

ἡμιόλιος means a whole plus a half, the ratio 3:2. It does not mean 'one half'. ὅπερ ἔδει δειχθῆναι is the closing formula 'which was to be shown'.

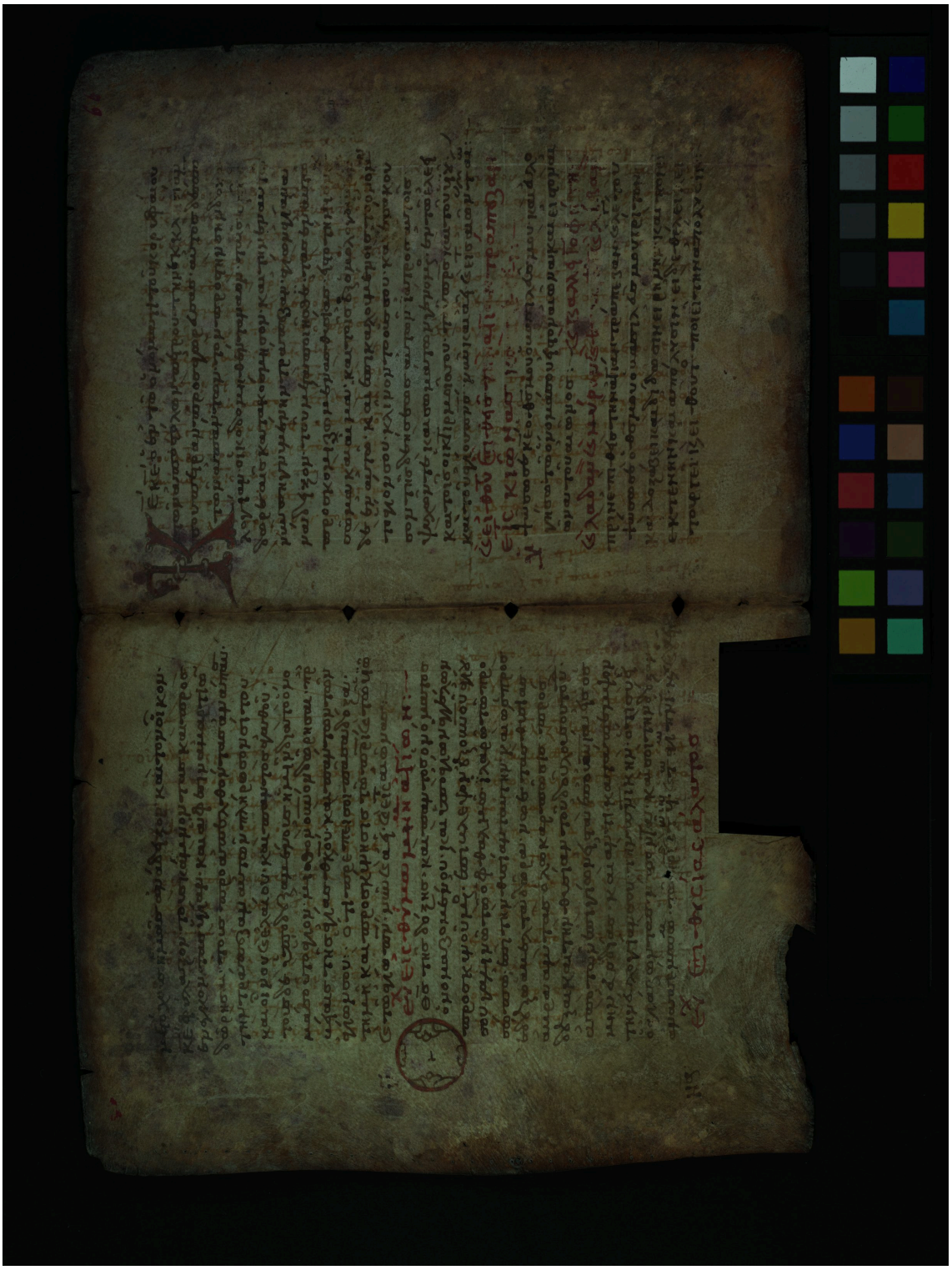
Here the comparison uses the cylinder that encloses the sphere: radius r, height 2r. It has one quarter the volume of the large auxiliary cylinder. Keeping those two cylinders distinct is essential. The mechanical discovery still has to be distinguished from a complete geometrical demonstration, as Archimedes explains before this argument.



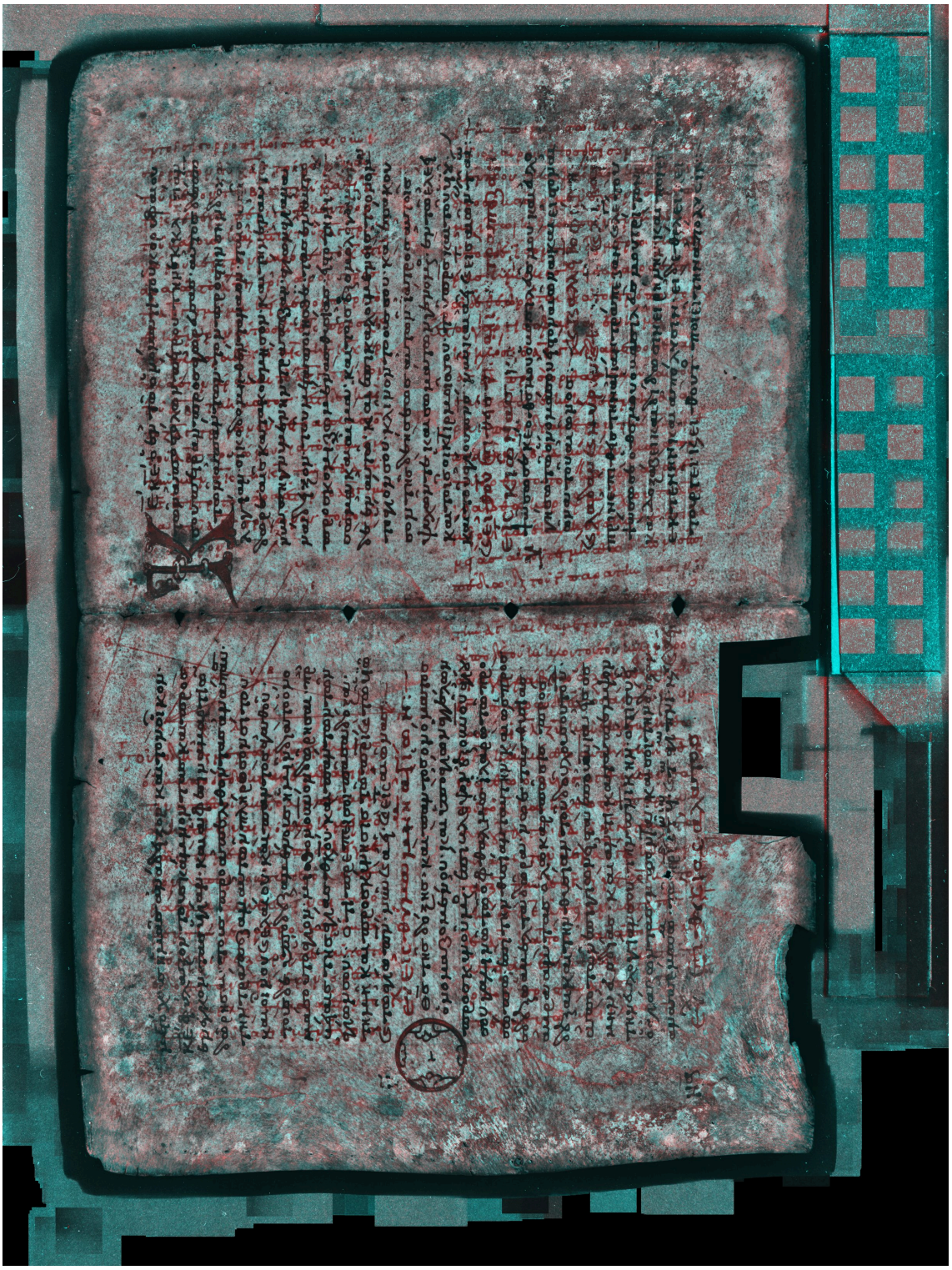
Mapped excerpt in the supplied processed image; project coordinates, not a new decipherment.

Arch17r: the two image products

Full frame, reduced from the project's 8160 × 10880 registered TIFF. Existing line coordinates locate selected excerpts; see the individual readings. No letter has been retouched.



Project true_pack8 visible composite.

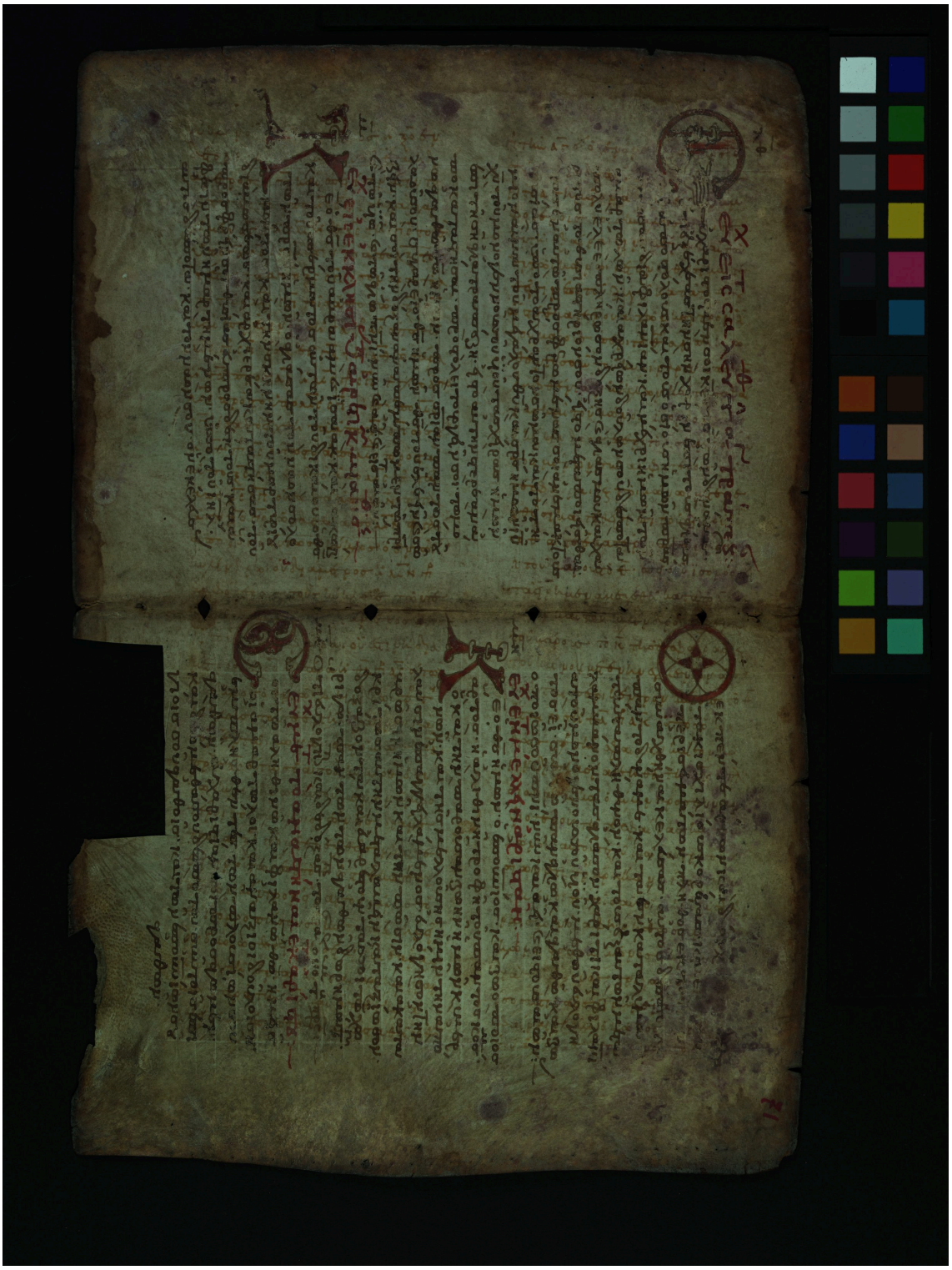


Project pseudo_no-veil processed image.

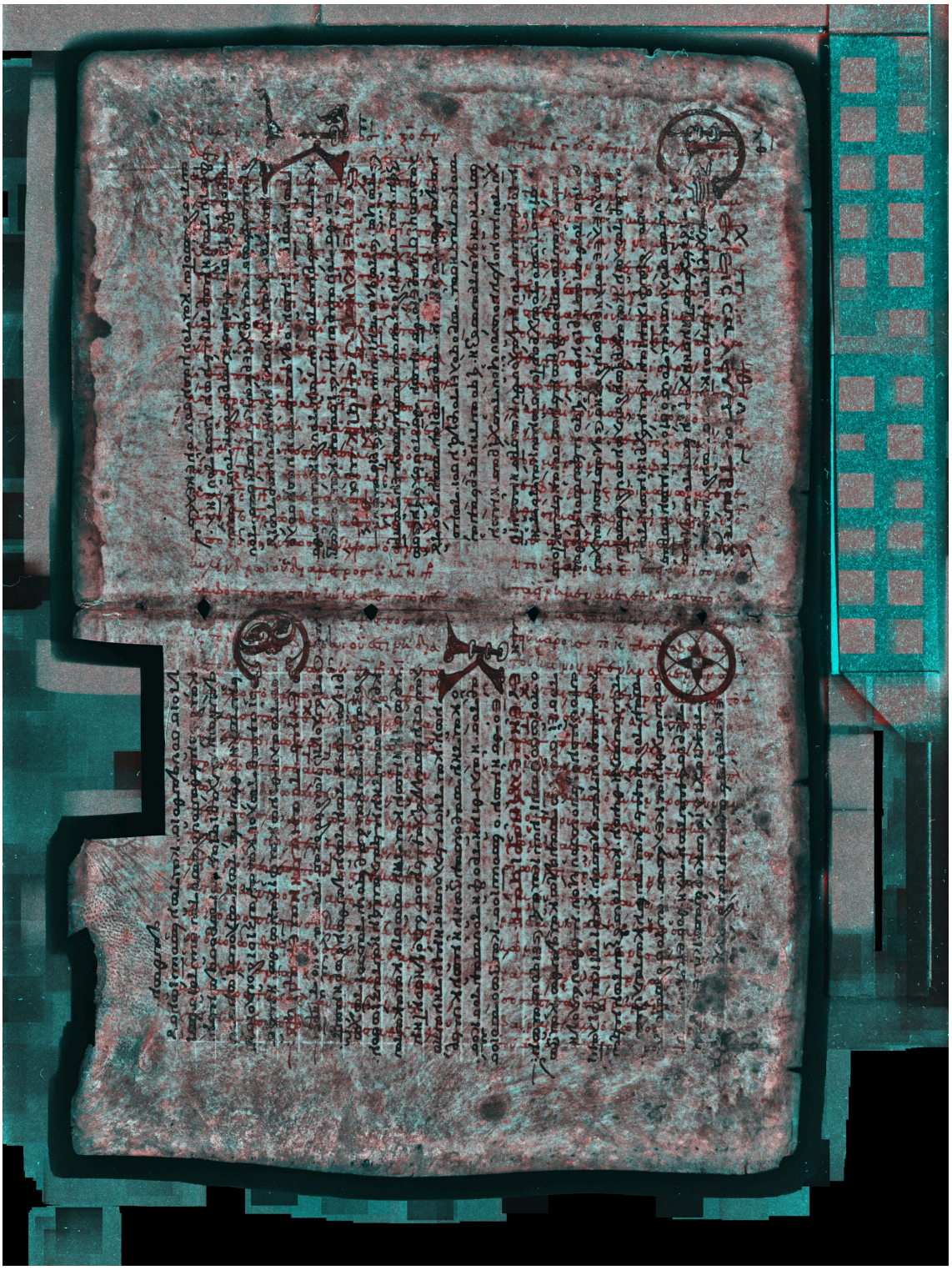
Original project files and full TIFFs

Arch17v: the two image products

Full frame, reduced from the project's 8160 × 10880 registered TIFF. Existing line coordinates locate selected excerpts; see the individual readings. No letter has been retouched.



Project true_pack8 visible composite.

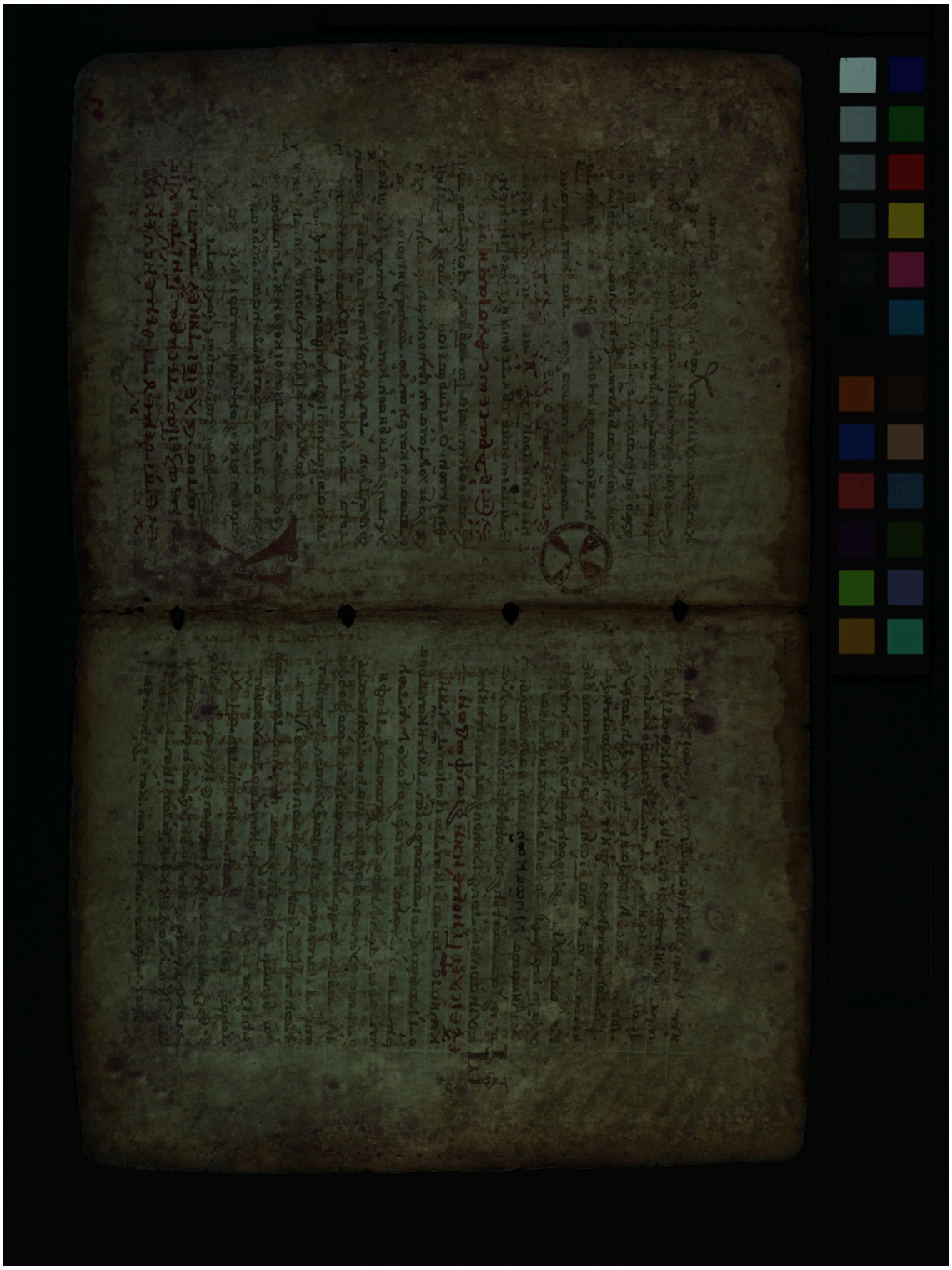


Project pseudo_no-veil processed image.

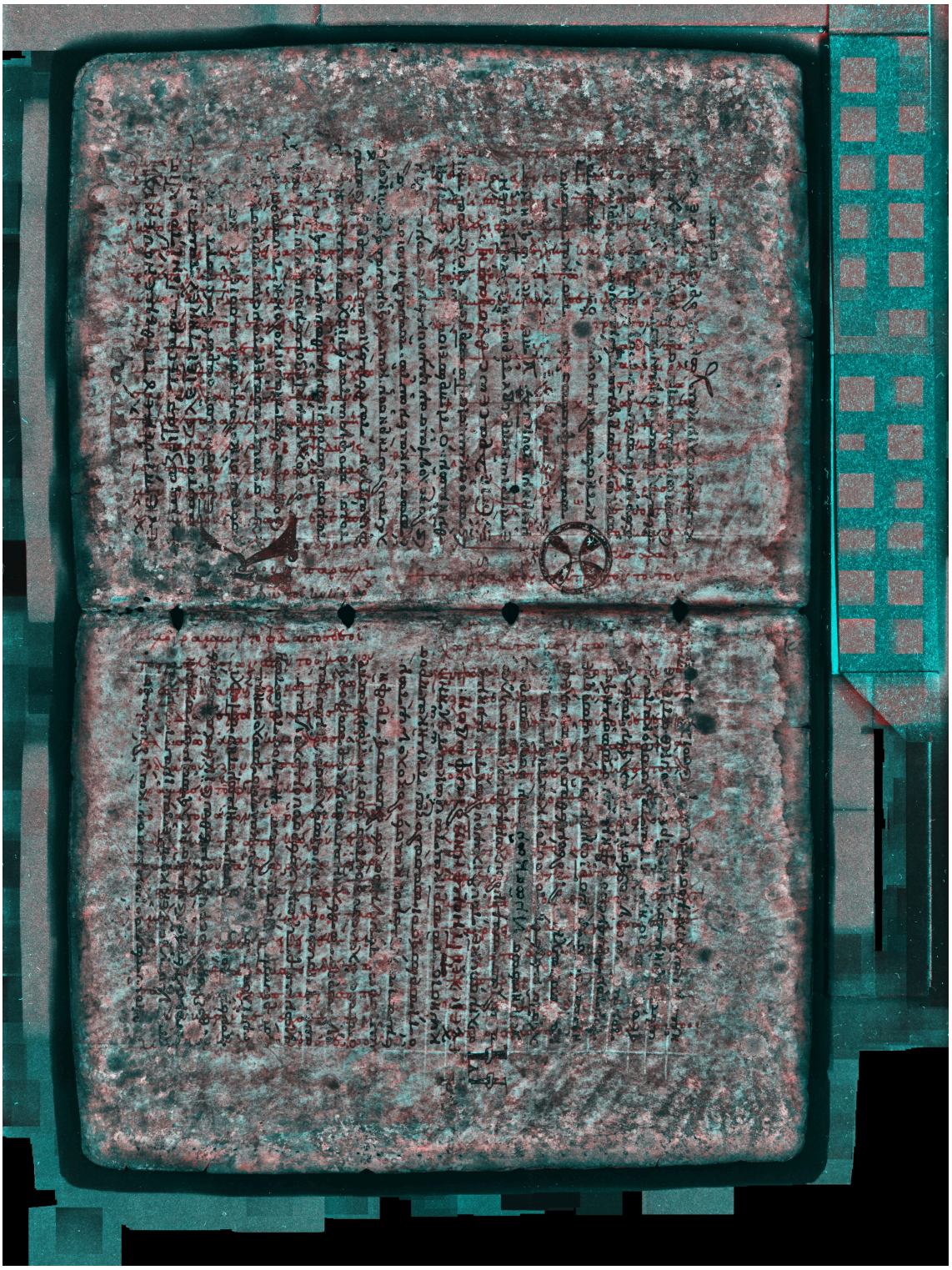
Original project files and full TIFFs

Arch18r: the two image products

Full frame, reduced from the project's 8160 × 10880 registered TIFF. Existing line coordinates locate selected excerpts; see the individual readings. No letter has been retouched.



Project true_pack8 visible composite.



Project pseudo_no-veil processed image.

Original project files and full TIFFs

Retained source files

Unmodified transcriptions and image provenance can be saved directly from this self-contained file.

- 065r-072v_Arch18r_TEI_Netz-Wilson.xml
- 066r-071v_Arch17r_TEI_Netz-Wilson.xml
- 066v-071r_Arch17v_TEI_Netz-Wilson.xml
- 0_ReadMe.txt
- Arch17r-pseudo_no-veil-metadata.txt
- Arch17r-true_pack8-metadata.txt
- Arch17v-pseudo_no-veil-metadata.txt
- Arch17v-true_pack8-metadata.txt
- Arch18r-pseudo_no-veil-metadata.txt
- Arch18r-true_pack8-metadata.txt
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- CODE-LICENSE.txt
- FolioIndex.txt
- Method-Heiberg-p5.xml
- Method-NW-p5.xml
- NOTICES.txt
- TEI_Encoding.txt
- images.json

Credits and redistribution

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0_ReadMe.txt, encoding documentation and FolioIndex.txt are included unchanged.

We retain unmodified Method-NW-p5.xml, Method-Heiberg-p5.xml, and the mapped
Netz-Wilson XML for Arch17r, Arch17v and Arch18r. The Heiberg file reconstructs
his reading of Codex C from the apparatus; it is NOT his complete critical

edition. The digital release incorporates updates described in its 2011 notes.

ADAPTATIONS
Ten selected Greek excerpts use the expanded side of TEI choice, join split words and omit marginal/superscript markers and figure labels. Supplied letters have red angle brackets, uncertain readings dotted underlines and expanded abbreviation letters italic. Word spacing, accents and punctuation are editorial. The reading does not silently correct the earlier 'twice' to the later 'four times'.

Six images are full-frame JPEG reductions of the project's registered TIFFs: Sinar_true_pack8 and Sinar_pseudo_no-veil for the three folios. images.json records original URLs, TIFF hashes, dimensions and exact reduction parameters. Selection rectangles reuse the project's line coordinates. They are not new recognition, retouched letters, or claims of an independently deciphered reading. The visible composite and processed image are distinct supplied project products.

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MATHEMATICAL CROSS-CHECK
T. L. Heath, The Method of Archimedes, recently discovered by Heiberg (Cambridge, 1912), proposition 2, printed pp. 18-21. Public-domain English translation.
https://www.wilbourhall.org/pdfs/archimedes/archimedesHeiberg.pdf
Our new English and Latin are not presented as Heath's words. The source labels follow the Greek transcription, including Theta (Theta is often H in redrawings).

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